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Battery pack comprising extinguishment unit

Abstract

A battery pack includes at least two battery modules arranged in one direction; and a fire extinguisher having a linear temperature sensor partially extending linearly along the at least two battery modules and configured to sense whether at least one of the at least two battery modules has a temperature over a predetermined temperature, a fire extinguishing tank configured to accommodate a fire extinguishing agent therein, a pipe connected to the fire extinguishing tank to supply the fire extinguishing agent from the fire extinguishing tank to each of the at least two battery modules, and a valve opened to supply the fire extinguishing agent from the fire extinguishing tank to the battery module over the predetermined temperature.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a Continuation of U.S. application Ser. No. 17/628,055, filed on Jan. 18, 2022, which is the National Phase of PCT/KR2020/010509, filed on Aug. 7, 2020, which claims priority under 35 U.S.C. 119(a) to Korean Patent Application No. 10-2019-0096939, filed on Aug. 8, 2019 in the Republic of Korea, all of which are hereby expressly incorporated by reference in their entirety into the present application.

TECHNICAL FIELD

(1) The present disclosure relates to a battery pack including a fire extinguishing unit, and more particularly, to a battery pack having a reduced risk of secondary ignition or explosion.

BACKGROUND ART

(2) Secondary batteries commercialized at the present include nickel cadmium batteries, nickel hydrogen batteries, nickel zinc batteries, and lithium secondary batteries. Among them, lithium secondary batteries are in the spotlight due to free advantages such as free charging and discharging by little memory effect compared to nickel-based secondary batteries, and very low self-discharge rate and high energy density.

(3) The lithium secondary battery mainly uses a lithium-based oxide and a carbon material as a positive electrode active material and a negative electrode active material, respectively. The lithium secondary battery includes an electrode assembly in which a positive electrode plate and a negative electrode plate respectively coated with the positive electrode active material and the negative electrode active material are disposed with a separator being interposed therebetween, and an exterior, namely a battery pouch exterior, for hermetically storing the electrode assembly together with an electrolyte.

(4) Recently, secondary batteries are widely used not only in small devices such as portable

electronic devices, but also in middle-sized or large-sized devices such as vehicles and energy storage systems. When used in the middle-sized or large-sized device, a large number of secondary batteries are electrically connected to increase capacity and output. In particular, pouch-type secondary batteries are widely used in the middle-sized or large-sized devices since they may be easily stacked.

(5) Meanwhile, recently, as the need for a large-capacity structure increases along with the use as an energy storage source, the demand for a battery pack including a plurality of secondary batteries electrically connected in series and/or in parallel, and a battery module accommodating the secondary batteries therein, and a battery management system (BMS) is increasing.

(6) In addition, the battery pack generally includes an outer housing made of a metal material to protect or store the plurality of secondary batteries from an external shock. Meanwhile, the demand for high-capacity battery packs is increasing recently.

(7) However, since the conventional battery pack or the conventional battery rack has a plurality of battery modules, if the secondary batteries of each battery module generates thermal runaway to cause ignition or explosion, heat or flame may be transferred to neighboring secondary batteries to cause secondary explosions, so efforts to prevent secondary ignition or explosion are increasing.

(8) Accordingly, it is necessary to develop a fast and complete fire extinguishing technology to take immediate action when thermal runaway occurs in some secondary batteries in the battery pack or the battery rack.

DISCLOSURE

Technical Problem

(9) The present disclosure is designed to solve the problems of the related art, and therefore the present disclosure is directed to providing a battery pack, which reduces the risk of secondary ignition or explosion.

(10) These and other objects and advantages of the present disclosure may be understood from the following detailed description and will become more fully apparent from the exemplary embodiments of the present disclosure. Also, it will be easily understood that the objects and advantages of the present disclosure may be realized by the means shown in the appended claims and combinations thereof.

Technical Solution

(11) In one aspect of the present disclosure, there is provided a battery pack, comprising: at least two battery modules arranged in one direction; and a fire extinguisher having a linear temperature sensor partially extending linearly along the at least two battery modules and configured to sense whether at least one of the at least two battery modules has a temperature over a predetermined temperature, a fire extinguishing tank configured to accommodate a fire extinguishing agent, a pipe connected to the fire extinguishing tank to supply the fire extinguishing agent from the fire extinguishing tank to each of the at least two battery modules, and a valve to supply the fire extinguishing agent from the fire extinguishing tank to the at least one battery module having a temperature over the predetermined temperature.

(12) Also, each of the at least two battery modules may have a gas passage configured to discharge a gas generated therein to the outside, and a plurality of gas discharge holes formed at an end of the gas passage to communicate with the outside.

(13) A part of the linear temperature sensor may be located at an outer side of the two or more battery modules to face the gas discharge holes respectively provided to the at least two battery modules.

(14) Moreover, the pipe may include a common pipe connected to an outlet hole of the fire extinguishing tank through which the fire extinguishing agent is output, and a distribution pipe having a distributed structure to be connected from the common pipe to an inlet hole of each of the at least two battery modules through which the fire extinguishing agent is input.

(15) In addition, the valve may include a common valve located in the common pipe to open or

close the common pipe, and a distribution valve located in the distribution pipe to open or close the distribution pipe.

(16) Further, the distribution valve may be a passive valve configured to be opened to inject the fire extinguishing agent into the battery module over the predetermined temperature.

(17) Also, at least a part of the passive valve may be inserted into the inlet hole, wherein the inlet hole is formed to communicate with the gas passage provided to the battery module.

(18) In addition, the common valve may be an active valve configured to be opened when the linear temperature sensor senses the at least one battery module having a temperature over the predetermined temperature.

(19) Further, the fire extinguisher may further include a location calculating part configured to receive a signal from the linear temperature sensor and calculate a location of the at least one battery module having a temperature over the predetermined temperature.

(20) Also, each of the at least two battery modules may have two or more inlet holes.

(21) Moreover, the distribution pipe may be two distribution pipes respectively connected to the two or more inlet holes.

(22) In addition, a first distribution pipe of the two distribution pipes may include a passive valve configured to be opened when the at least one battery module is heated over the predetermined temperature.

(23) Further, a second distribution pipe of the two distribution pipes may include an active valve configured to be opened when the at least one battery module is heated over the predetermined temperature.

(24) Also, the fire extinguisher may further include a smoke sensor configured to sense a smoke discharged from the at least two battery modules.

(25) Moreover, in another aspect of the present disclosure, there is also provided a battery rack, comprising: a battery pack; and a rack case configured to accommodate the battery pack.

(26) In addition, in another aspect of the present disclosure, there is also provided an energy storage system, comprising two or more battery racks.

Advantageous Effects

(27) According to an embodiment of the present disclosure, since the battery pack includes the linear temperature sensor linearly extending along at least two battery modules, it is possible to reduce the manufacturing cost of the battery pack.

(28) That is, when a plurality of temperature sensors is provided in the prior art, a plurality of temperature sensors and separate signal wires for connecting the plurality of temperature sensors are required, which increases the manufacturing cost due to high material cost and long installation work. Meanwhile, the battery pack of the present disclosure uses only one linear temperature sensor to detect the temperature of the plurality of battery modules, so a separate signal wire is not required and easy installation is secured due to a light and flexible design. Thus, the manufacturing cost of battery pack may be greatly reduced.

(29) Moreover, the linear temperature sensor is useful for setting a plurality of points for more accurate temperature sensing even for one battery module. Accordingly, in the present disclosure, it is possible to greatly reduce the failure rate in detecting the occurrence of fire in the battery module.

(30) In addition, according to an embodiment of the present disclosure, if a part of the linear temperature sensor is located at the outer side of the two or more battery modules to face the gas discharge hole provided to each of the at least two battery modules, when thermal runaway occurs at the cell assembly of the battery module, hot air with elevated temperature is discharged through the gas discharge hole or hot gas generated during ignition of the cell assembly is discharged through the gas discharge hole, so the linear temperature sensor may sense the hot air or gas with fast response. Accordingly, it is possible to increase the safety by rapidly responding to the thermal runaway of the battery pack according to the present disclosure.

(31) In addition, according to an embodiment of the present disclosure, since at least a part of the passive valve is inserted into the inlet hole perforated to communicate with the gas discharge passage provided to the battery module, when thermal runaway occurs, the passive valve opens so that the fire extinguishing agent may be injected individually only to the battery module at which the thermal runaway occurs. Moreover, since the fire extinguishing agent may be injected directly into the battery module, rather than to an outside thereof, it is possible to effectively extinguish and cool the fire of the battery module at which thermal runaway occurs.

Description

DESCRIPTION OF DRAWINGS

- (1) The accompanying drawings illustrate a preferred embodiment of the present disclosure and together with the foregoing disclosure, serve to provide further understanding of the technical features of the present disclosure, and thus, the present disclosure is not construed as being limited to the drawing.
- (2) FIG. 1 is a front perspective view schematically showing a battery pack according to an embodiment of the present disclosure.
- (3) FIG. 2 is a rear perspective view schematically showing the battery pack according to an embodiment of the present disclosure.
- (4) FIG. 3 is a diagram schematically showing components of the battery pack according to an embodiment of the present disclosure.
- (5) FIG. 4 is a diagram schematically showing some components of a fire extinguishing unit, employed at the battery pack according to an embodiment of the present disclosure.
- (6) FIG. 5 is a rear perspective view schematically showing a partial internal structure of a battery module, employed at the battery pack according to an embodiment of the present disclosure.
- (7) FIG. 6 is a rear perspective view schematically showing the battery module, employed at the battery pack according to an embodiment of the present disclosure.
- (8) FIG. 7 is a front perspective view schematically showing the battery module, employed at the battery pack according to an embodiment of the present disclosure.
- (9) FIG. 8 is a partial rear view schematically showing a portion of the battery pack according to an embodiment of the present disclosure.
- (10) FIG. 9 is a partial perspective view schematically showing a portion of the battery pack according to an embodiment of the present disclosure.
- (11) FIG. 10 is a sectional view schematically showing of an internal configuration of a portion of the battery pack according to an embodiment of the present disclosure.
- (12) FIG. 11 is a partial perspective view schematically showing a pipe and a valve, employed at the battery pack according to an embodiment of the present disclosure.
- (13) FIG. 12 is a rear perspective view schematically showing a battery module, employed at a battery pack according to another embodiment of the present disclosure.
- (14) FIG. 13 is a partial perspective view schematically showing a pipe and a valve, employed at the battery pack according to another embodiment of the present disclosure.
- (15) FIGS. 14 and 15 are a front view and a plan view schematically showing a smoke sensor, employed at the battery pack according to an embodiment of the present disclosure.
- (16) FIG. 16 is a front view schematically showing an energy storage system according to an embodiment of the present disclosure.

BEST MODE

- (17) Hereinafter, preferred embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. Prior to the description, it should be understood that the terms used in the specification and the appended claims should not be construed as limited to

general and dictionary meanings, but interpreted based on the meanings and concepts corresponding to technical aspects of the present disclosure on the basis of the principle that the inventor is allowed to define terms appropriately for the best explanation.

(18) Therefore, the description proposed herein is just a preferable example for the purpose of illustrations only, not intended to limit the scope of the disclosure, so it should be understood that other equivalents and modifications could be made thereto without departing from the scope of the disclosure.

(19) FIG. 1 is a front perspective view schematically showing a battery pack according to an embodiment of the present disclosure. FIG. 2 is a rear perspective view schematically showing the battery pack according to an embodiment of the present disclosure. Also, FIG. 3 is a diagram schematically showing components of the battery pack according to an embodiment of the present disclosure.

(20) Referring to FIGS. 1 to 3, a battery pack **400** according to an embodiment of the present disclosure includes at least two battery modules **200** arranged in one direction, and a fire extinguishing unit **300** configured to extinguish a fire of the battery module **200**.

(21) Specifically, the fire extinguishing unit **300** may include a linear temperature sensor **310**, a fire extinguishing tank **320**, a pipe **330**, and a valve **340**.

(22) First, the fire extinguishing tank **320** may contain a fire extinguishing agent (not shown) therein. For example, the fire extinguishing agent may be a concentrated solution of an inorganic salt such as potassium carbonate, a chemical bubble, an air bubble, carbon dioxide, or water. In addition, the fire extinguishing tank **320** may have a compressed gas therein to inject or move the fire extinguishing agent at an appropriate pressure along the pipe **330**. For example, the capacity of the fire extinguishing tank **320** may be 59 L, the compressed gas may be nitrogen of 8 bar, and the fire extinguishing agent may be 40 L of water. Here, if the fire extinguishing agent is water, when the fire extinguishing agent is sprayed into the battery module **200**, the fire extinguishing agent has a heat shielding effect together with the fire extinguishing and cooling effect, so it is effective in preventing thermal propagation when high-temperature gas and flame are generated due to thermal runaway. As a result, it is possible to effectively prevent a fire or thermal runaway from propagating among the plurality of battery modules **200**.

(23) The pipe **330** may be configured to be connected to supply the fire extinguishing agent to each of the at least two battery modules **200**. For example, the pipe **330** may be made of a material that is not corroded by water. For example, the pipe **330** may be made of stainless steel. One end of the pipe **330** may be connected to an outlet hole **321** of the fire extinguishing tank **320**. The other end of the pipe **330** may have a shape extending to the inside of each of the at least two battery modules **200**.

(24) For example, the pipe **330** may include a common pipe **333** connected to the outlet hole **321** of the fire extinguishing tank **320** through which the fire extinguishing agent is discharged, a distribution pipe **336** having a distributed structure to be connected to an inlet hole **264** provided in each of the at least two battery modules **200** from the common pipe **333**. For example, as shown in FIG. 2, the pipe **330** may include one common pipe **333** connected to the outlet hole **321** of the fire extinguishing tank **320**, and eight distribution pipes **336** branched from the common pipe **333**. In addition, the eight distribution pipes **336** may be configured to be connected to the inlet holes **264** of eight battery modules **200**.

(25) In addition, the valve **340** may be configured to be opened to supply the fire extinguishing agent from the fire extinguishing tank **320** to a battery module **200** over a predetermined temperature through the pipe **330**. Specifically, the valve **340** may be an active valve capable of controlling the opening and closing of the valve **340** by receiving a signal from the fire extinguishing unit **300**. More specifically, the active valve may be a control valve, a motor-operated valve, a solenoid valve, or a pneumatic valve.

(26) In addition, the linear temperature sensor **310** may be configured to sense whether at least one

of the at least two battery modules **200** has a temperature over the predetermined temperature.

(27) For example, the linear temperature sensor **310** may be configured to melt when a heat sensing material coated on two wires reaches a temperature higher over a reference temperature, to cause a short circuit between the two wires, thereby emitting a fire or overheat signal. For example, the heat sensing material may be a thermoplastic resin that melts at 70° C. to 100° C. For example, the thermoplastic resin may be a polyester resin or an acrylic resin. Additionally, the linear temperature sensor **310** may further include an insulating coating material configured to surround the heat sensing material. The coating material may include polyvinyl chloride.

(28) In addition, the linear temperature sensor **310** may have a structure extending linearly along at least two battery modules **200** arranged in one direction. For example, as shown in FIG. 2, the battery pack **400** may include eight battery modules **200** arranged in a vertical direction. The linear temperature sensor **310** may be configured so that one end thereof is connected to the controller **350** and extends downward along the eight battery modules **200** arranged in the vertical direction, and the other end thereof is connected to a resistor **315** at a distal end. At this time, a bracket (not shown) and a fixing buckle (not shown) may be used to partially fix the position of the linear temperature sensor **310**.

(29) Therefore, according to this configuration of the present disclosure, since the battery pack **400** includes the linear temperature sensor **310** linearly extending along at least two battery modules **200**, it is possible to reduce the manufacturing cost of the battery pack.

(30) That is, when a plurality of temperature sensors is provided in the prior art, a plurality of temperature sensors and separate signal wires for connecting the plurality of temperature sensors are required, which increases the manufacturing cost due to high material cost and long installation work. Meanwhile, the battery pack **400** of the present disclosure uses only one linear temperature sensor **310** to detect the temperature of the plurality of battery modules **200**, so a separate signal wire is not required and easy installation is secured due to a light and flexible design. Thus, the manufacturing cost of battery pack **400** may be greatly reduced.

(31) Moreover, the linear temperature sensor **310** is useful for setting a plurality of points for more accurate temperature sensing even for one battery module **200**. Accordingly, in the present disclosure, it is possible to greatly reduce the failure rate in detecting the occurrence of fire in the battery module **200**.

(32) FIG. 4 is a diagram schematically showing some components of a fire extinguishing unit, employed at the battery pack according to an embodiment of the present disclosure.

(33) Referring to FIG. 4 along with FIG. 3, the fire extinguishing unit **300** may include a controller **350**. More specifically, the controller **350** may include a valve opening and closing part **353** and a location calculating part **356**.

(34) Specifically, the valve opening and closing part **353** may be configured to open the valve **340** when the linear temperature sensor **310** senses a temperature over the predetermined temperature. For example, when the valve **340** is an active valve, the valve opening and closing part **353** may be configured to transmit a signal for controlling the active valve.

(35) In addition, the location calculating part **356** may be configured to receive a signal from the linear temperature sensor **310** and calculate a location of the battery module **200** over the predetermined temperature. More specifically, the location calculating part **356** may analyze the signal received from the linear temperature sensor **310**. In addition, the location calculating part **356** may calculate the location to identify which battery module **200** among the at least two battery modules **200** is heated over the predetermined temperature.

(36) For example, if a temperature over the predetermined temperature is detected in a part of the entire length of the linear temperature sensor **310**, the location calculating part **356** may estimate the battery module **200** at which thermal runaway occurs by calculating the distance between the detected part of the linear temperature sensor **310** and the fire extinguishing unit **300**.

(37) FIG. 5 is a rear perspective view schematically showing a partial internal structure of a battery

module, employed at the battery pack according to an embodiment of the present disclosure.

(38) Referring to FIG. 5, the battery module **200** according to an embodiment of the present disclosure may include at least two cell assemblies **100** and a module housing **210**.

(39) Each of the at least two cell assemblies **100** may include a plurality of secondary batteries **110** stacked in a front and rear direction. The secondary battery **110** may be a pouch-type secondary battery **110**. For example, as shown in FIG. 5, when viewed in the F direction of FIG. 1 (from the front), each of the two cell assemblies **100** may be configured such that a plurality of pouch-type secondary batteries **110** are stacked side by side in the front and rear direction.

(40) Meanwhile, in this specification, unless otherwise specified, the upper, lower, front, rear, left and right directions will be set based on when viewed in the F direction.

(41) In particular, the pouch-type secondary battery **110** may include an electrode assembly (not shown), an electrolyte solution (not shown), and a pouch **116**.

(42) Moreover, a positive electrode lead **111** and a negative electrode lead (not shown) may be formed at left and right ends of the secondary battery **110**, which are opposite to each other based on the center of the secondary battery **110**. That is, the positive electrode lead **111** may be provided at one end of the secondary battery **110** based on the center thereof. In addition, the negative electrode lead may be provided at the other end of the secondary battery **110** based on the center thereof.

(43) However, the battery module **200** according to the present disclosure is not limited to the pouch-type secondary battery **110** described above, and various kinds of secondary batteries known at the time of filing of this application may be employed.

(44) Meanwhile, referring to FIG. 5 again, the battery module **200** may further include a bus bar assembly **270**. Specifically, the bus bar assembly **270** may include at least one bus bar **272** configured to electrically connect the plurality of secondary batteries **110** to each other and at least two bus bar frame **276** configured to mount the at least one bus bar **272** at an outer side. The at least two bus bar frame **276** may be provided at left and right sides of the cell assembly **100**, respectively.

(45) Meanwhile, the module housing **210** may have an inner space to accommodate the cell assembly **100** therein. Specifically, when viewed directly in the F direction of FIG. 1, the module housing **210** may include an upper cover **220**, a base plate **240**, a front cover **260**, and a rear cover **250**.

(46) Specifically, the base plate **240** may have an area larger than the size of a bottom surface of the at least two cell assemblies **100** so as to mount the at least two cell assemblies **100** to an upper portion thereof. The base plate **240** may have a plate shape extending in a horizontal direction.

(47) Here, the horizontal direction may refer to a direction parallel to the ground when the battery module **200** is placed on the ground, and may also refer to at least one direction on a plane perpendicular to the upper and lower direction.

(48) In addition, the upper cover **220** may include an upper wall **224** and a sidewall **226** extending downward from the upper wall **224**. The upper wall **224** may have a plate shape extending in a horizontal direction to cover an upper portion of the cell assembly **100**. The sidewall **226** may have a plate shape extending downward from both left and right ends of the upper wall **224** to cover both left and right sides of the cell assembly **100**.

(49) In addition, the sidewall **226** may be coupled to a portion of the base plate **240**. For example, as shown in FIG. 5, the upper cover **220** may include an upper wall **224** having a plate shape extending in the front, rear, left and right directions. The upper cover **220** may include two sidewalls **226** extending downward from both left and right ends of the upper wall **224**, respectively. Further, lower ends of the two sidewalls **226** may be configured to be coupled with both left and right ends of the base plate **240**, respectively. In this case, the coupling method may be a male and female coupling method or a welding method.

(50) Moreover, the front cover **260** may be configured to cover the front side of the plurality of

secondary batteries **110**. For example, the front cover **260** may have a plate shape larger than the size of the front surface of the plurality of secondary batteries **110**. The plate shape may be erected in a vertical direction.

(51) In addition, the rear cover **250** may be configured to cover the rear side of the cell assembly **100**. For example, the rear cover **250** may have a plate shape larger than the size of the rear surface of the plurality of secondary batteries **110**.

(52) Moreover, the module housing **210** may have a gas passage **211** through which the gas generated from the cell assembly **100** flows. Here, the gas passage **211** may be a space elongated in the front and rear direction to communicate with the outside. The gas passage **211** may be provided at one of the left and right sides or both left and right sides of the cell assembly **100**.

(53) More specifically, the gas passage **211** may be a space between the upper or lower portion of the cell assembly **100** and the module housing **210**. That is, the gas generated from the cell assembly **100** accommodated in the battery module **200** may move to both left and right sides of the cell assembly **100** through the gas passage **211** located at the upper or lower portion of the cell assembly **100** and be discharged out through a plurality of gas discharge holes **212** formed at the end of the gas passage **211** and perforated to communicate with the outside of the battery module **200**.

(54) FIG. **6** is a rear perspective view schematically showing the battery module, employed at the battery pack according to an embodiment of the present disclosure. Also, FIG. **7** is a front perspective view schematically showing the battery module, employed at the battery pack according to an embodiment of the present disclosure.

(55) Referring to FIGS. **6** and **7** along with FIG. **5**, an inlet hole **264** may be provided at the rear cover **250** located at the rear side of each of the at least two battery modules **200** so that the fire extinguishing agent is introduced therethrough. The inlet hole **264** may be positioned to communicate with the gas passage **211**. That is, the inlet hole **264** may be configured to communicate with the gas passages **211** located on both left and right sides based on the cell assembly **100**.

(56) For example, when viewed in the R direction of FIG. **6**, the inlet hole **264** may be provided at the right side of the rear cover **250**. In addition, the fire extinguishing agent introduced through the inlet hole **264** may move along the gas passage **211** located at the left side of the cell assembly **100**, and subsequently, the fire extinguishing agent may move to the gas passage **211** located at the right side of the cell assembly **100** through the gas passage **211** located at the upper or lower portion of the cell assembly **100**. Through this process, it is possible to extinguish and cool the ignited or overheated cell assembly **100** inside the battery module **200**.

(57) FIG. **8** is a partial rear view schematically showing a portion of the battery pack according to an embodiment of the present disclosure.

(58) Referring back to FIG. **8** along with FIG. **5**, the linear temperature sensor **310** may be positioned so that a part of the linear temperature sensor **310** faces the gas discharge hole **212** provided to each of the at least two battery modules **200**. For example, as shown in FIG. **8**, the gas discharge hole **212** may be provided to the rear cover **250** of each of the two or more battery modules **200** stacked in the vertical direction. In addition, the linear temperature sensor **310** may be disposed at the outer side of each of the two or more battery modules **200** to face the gas discharge hole **212**.

(59) Therefore, according to this configuration of the present disclosure, if a part of the linear temperature sensor **310** is located at the outer side of the two or more battery modules **200** to face the gas discharge hole **212** provided to each of the at least two battery modules **200**, when thermal runaway occurs at the cell assembly **100** of the battery module **200**, hot air with elevated temperature is discharged through the gas discharge hole **212** or hot gas generated during ignition of the cell assembly is discharged through the gas discharge hole **212**, so the linear temperature sensor **310** may sense the hot air or gas with fast response. Accordingly, it is possible to increase

the stability by rapidly responding to the thermal runaway of the battery pack **400** according to the present disclosure.

(60) FIG. **9** is a partial perspective view schematically showing a portion of the battery pack according to an embodiment of the present disclosure.

(61) Referring to FIG. **9** along with FIG. **7**, the valve **340** may include a common valve **343** located at a part of the common pipe **333** to open or close the common pipe **333**, and a distribution valve **346** located at a part of the distribution pipe **336** to open or close the distribution pipe **336**.

(62) For example, the common valve **343** may be an active valve capable of controlling the opening and closing of the valve **340** by receiving a signal from the valve opening and closing part of the fire extinguishing unit **300**. For example, when at least one of the at least two battery modules **200** is heated over the predetermined temperature, the temperature is detected by the location calculating part **356** of the fire extinguishing unit **300**, and the valve opening and closing part **353** may transmit an open signal to the common valve **343** to open the common valve **343** so that the extinguishing agent may be discharged from the fire extinguishing tank **320**. For example, the active valve may be a control valve, a motor-operated valve, a solenoid valve, or a pneumatic valve.

(63) FIG. **10** is a sectional view schematically showing of an internal configuration of a portion of the battery pack according to an embodiment of the present disclosure.

(64) Referring to FIG. **10** along with FIG. **8**, the distribution valve **346** may be a passive valve **347** configured to be opened so that the fire extinguishing agent may be injected into the battery module **200** over the predetermined temperature. For example, the passive valve **347** may be configured to open the valve **340** when the internal temperature of the battery module **200** is over the predetermined temperature. For example, the passive valve **347** may include a glass bulb **347a** and a predetermined fluid (not shown) accommodated in the glass bulb **347a**. The glass bulb **347a** may be configured to seal a passage **347c** of the passive valve **347** through which the fluid flows.

(65) In addition, the glass bulb **347a** may be configured to be broken by volume expansion of the predetermined liquid over the predetermined temperature, for example 70° C. to 100° C. That is, if the passive valve **347** is located inside the battery module **200**, when the internal temperature of the battery module **200** rises over the predetermined temperature, the glass bulb **347a** blocking the passage **347c** of the valve **340** through which the fluid flows may be broken open the passage **347c** of the valve. Further, the passive valve **347** may further include a sprinkler head **347b** having a dispersion pin to disperse the discharged fluid in all directions.

(66) FIG. **11** is a partial perspective view schematically showing a pipe and a valve, employed at the battery pack according to an embodiment of the present disclosure.

(67) Referring to FIG. **11** along with FIGS. **5** and **8**, the passive valve **347** may be configured to be at least partially inserted into the inlet hole **264** configured to communicate with the gas passage **211** provided to the battery module **200**. For example, the pipe **330** may include a plurality of distribution pipes **336** along the common pipe **333**, and the distribution pipe **336** may be a Tee pipe or an elbow pipe. In addition, the distribution valve **346** may have provided at the distal end of the distribution pipe **336**. The distribution valve **346** may be the passive valve **347**. For example, the passive valve **347** may include a glass bulb and a sprinkler head. In addition, at least a part of the distribution valve **346** may be inserted into the inlet hole **264** configured to communicate with the gas passage **211** provided to the battery module **200**.

(68) Therefore, according to this configuration of the present disclosure, since at least a part of the passive valve **347** is inserted into the inlet hole **264** perforated to communicate with the gas discharge passage provided to the battery module **200**, when thermal runaway occurs, the passive valve **347** opens so that the fire extinguishing agent may be injected individually only to the battery module at which the thermal runaway occurs. Moreover, since the fire extinguishing agent may be injected directly into the battery module **200**, rather than to an outside thereof, it is possible to effectively extinguish and cool the fire of the battery module **200** at which thermal runaway occurs.

(69) FIG. 12 is a rear perspective view schematically showing a battery module, employed at a battery pack according to another embodiment of the present disclosure. Also, FIG. 13 is a partial perspective view schematically showing a pipe and a valve, employed at the battery pack according to another embodiment of the present disclosure.

(70) Referring to FIGS. 12 and 13 along with FIGS. 3 and 4, a battery module 200A of another embodiment may have two or more inlet holes 264. A distribution pipe 336A may be connected to each of the two or more inlet holes 264. Any one of the two distribution pipes 336A may include the passive valve 347 that is opened when the battery module 200A is heated over the predetermined temperature. The passive valve 347 may be configured to be opened so that the fire extinguishing agent may be injected into the battery module 200A over the predetermined temperature. For example, the passive valve 347 may be configured to be opened when the internal temperature of the battery module 200A increases over the predetermined temperature. The passive valve 347 may be located at the distal end of the distribution pipe 336A so that at least a part of the passive valve 347 may be inserted into the inlet hole 264.

(71) In addition, as shown in FIG. 13, a pipe 330A of the battery pack according to another embodiment includes a common pipe 333A and at least two or more distribution pipes 336A connected to the common pipe 333A. At this time, one of the two distribution pipes 336A may be connected to the passive valve 347, and the other of the two distribution pipes 336A may be connected to the active valve 348 that is opened when the battery module 200A is heated over the predetermined temperature.

(72) The active valve 348 may be configured to be opened when the linear temperature sensor 310 senses a battery module 200A over the predetermined temperature. For example, as shown in FIGS. 4 and 13, the active valve 348 may be configured to operate by receiving an opening and closing signal from the valve opening and closing part 353 (FIG. 4) of the controller 350 of the fire extinguishing unit 300. The active valve 348 may be, for example, a motor valve or a solenoid valve.

(73) Therefore, according to this configuration of the present disclosure, since one of the two distribution pipes 336A includes the passive valve 347 opened when the battery module 200A is heated over the predetermined temperature and the other of the two distribution pipes 336A includes the active valve 348, even if the passive valve 347 is not opened when thermal runaway occurs at the battery module 200A, the fire extinguishing unit 300 may transmit a signal to the active valve 348 to be opened, thereby increasing the safety of the battery pack 400.

(74) FIGS. 14 and 15 are a front view and a plan view schematically showing a smoke sensor, employed at the battery pack according to an embodiment of the present disclosure.

(75) Referring to FIGS. 14 and 15 along with FIG. 3, the fire extinguishing unit 300 may further include a smoke sensor 370 configured to sense a smoke discharged from the at least two battery modules 200. Specifically, the smoke sensor 370 may be located at an uppermost portion T (FIG. 1) of the at least two battery modules 200 stacked in the vertical direction. That is, if a fire occurs in the battery module 200, the generated gas may be moved upward, so it is preferable that the smoke sensor 370 is located at the uppermost portion T (FIG. 1) of the at least two battery modules 200.

(76) In addition, the smoke sensor 370 may be configured to transmit a signal to the controller 350 of the fire extinguishing unit 300 when detecting smoke. The controller 350 may allow the valve opening and closing part 353 to open the common valve 343 and/or the distribution valve 346 according to the received signal.

(77) FIG. 16 is a front view schematically showing an energy storage system according to an embodiment of the present disclosure.

(78) Referring to FIG. 16, a battery rack 500 according to an embodiment of the present disclosure may include the battery pack 400 and a rack case 510 for accommodating the battery pack 400. The rack case 510 may also be configured to accommodate the battery pack 400 in a state where a plurality of battery modules 200 are vertically stacked. Inside the rack case 510, the battery module

200 may be mounted such that its lower surface is in a parallel shape to the horizontal surface.

(79) Moreover, the rack case **510** is configured to have at least one side openable, and the battery module **200** may be inserted into the inner space through the open side. However, the rack case **510** may also be configured to allow such an open side to be closed.

(80) In addition, the battery rack **500** may further include a battery management system **530** (BMS) or the like.

(81) Meanwhile, an energy storage system **600** according to an embodiment of the present disclosure may include two or more battery racks **500**. The two or more battery racks **500** may be arranged in one direction. For example, as shown in FIG. **16**, the energy storage system **600** may be configured such that three battery racks **500** are arranged in one direction. In addition, the energy storage system **600** may have a central controller (not shown) capable of controlling charging and discharging of three battery racks **500**.

(82) Meanwhile, even though the terms indicating directions such as upper, lower, left, right, front and rear directions are used in the specification, it is obvious to those skilled in the art that these merely represent relative locations for convenience in explanation and may vary based on a location of an observer or an object.

(83) The present disclosure has been described in detail. However, it should be understood that the detailed description and specific examples, while indicating preferred embodiments of the disclosure, are given by way of illustration only, since various changes and modifications within the scope of the disclosure will become apparent to those skilled in the art from this detailed description.

REFERENCE SIGNS

(84) TABLE-US-00001 Reference Signs 200: battery module 100: cell assembly 110: secondary battery 210: module housing 211: gas passage 212: gas discharge hole 264: inlet hole 300: fire extinguishing unit 310: linear temperature sensor 320: fire extinguishing tank 321: outlet hole 330, 333, 336: pipe, common pipe, distribution pipe 340, 343, 346: valve, common valve, distribution valve 350, 353, 356: controller, valve opening and closing part, location calculating part 370: smoke sensor 400: battery pack 500: battery rack 510: rack case 600: energy storage system

INDUSTRIAL APPLICABILITY

(85) The present disclosure relates to a battery pack. In addition, the present disclosure is available for industries associated with a battery rack including the battery pack, and a large-scale energy storage system including a plurality of battery racks.

Claims

1. A battery pack, comprising: at least two battery modules arranged in one direction; a fire extinguisher; a fire extinguishing tank configured to accommodate a fire extinguishing agent; a pipe connected to the fire extinguishing tank to supply the fire extinguishing agent from the fire extinguishing tank to each of the at least two battery modules; a temperature sensor measuring the temperature of each of the at least two battery modules; and a valve to supply the fire extinguishing agent from the fire extinguishing tank to the at least one battery module having a temperature over a predetermined temperature, wherein the pipe includes a common pipe connected to an outlet hole of the fire extinguishing tank through which the fire extinguishing agent is output, and a distribution pipe having a distributed structure to be connected from the common pipe to an inlet hole of each of the at least two battery modules through which the fire extinguishing agent is input, wherein the valve includes a common valve located in the common pipe to open or close the common pipe, and a distribution valve located in the distribution pipe to open or close the distribution pipe, and wherein the distribution valve is a passive valve configured to be opened to inject the fire extinguishing agent into the battery module over the predetermined temperature.
2. The battery pack according to claim 1, wherein the temperature sensor is a linear temperature

sensor partially extending linearly along the at least two battery modules and configured to sense whether at least one of the at least two battery modules has a temperature over a predetermined temperature.

3. The battery pack according to claim 2, wherein each of the at least two battery modules has a gas passage configured to discharge a gas generated therein to the outside, and a plurality of gas discharge holes formed at an end of the gas passage to communicate with the outside, and wherein a part of the linear temperature sensor is located at an outer side of the two or more battery modules to face the gas discharge holes respectively provided to the at least two battery modules.

4. The battery pack according to claim 1, wherein at least a part of the passive valve is inserted into the inlet hole, and wherein the inlet hole is formed to communicate with the gas passage provided to the battery module.

5. The battery pack according to claim 2, wherein the common valve is an active valve configured to be opened when the linear temperature sensor senses the at least one battery module having a temperature over the predetermined temperature.

6. The battery pack according to claim 2, wherein the fire extinguisher further includes a location calculating part configured to receive a signal from the linear temperature sensor and calculate a location of the at least one battery module having a temperature over the predetermined temperature.

7. The battery pack according to claim 1, wherein each of the at least two battery modules has two inlet holes, wherein the distribution pipe is two distribution pipes respectively connected to the two inlet holes, wherein a first distribution pipe of the two distribution pipes includes a passive valve configured to be opened when the at least one battery module is heated over the predetermined temperature, and wherein a second distribution pipe of the two distribution pipes includes an active valve configured to be opened when the at least one battery module is heated over the predetermined temperature.

8. The battery pack according to claim 1, wherein the fire extinguisher further includes a smoke sensor configured to sense a smoke discharged from the at least two battery modules.

9. A battery rack, comprising: a battery pack according to claim 1; and a rack case configured to accommodate the battery pack.

10. An energy storage system, comprising two or more battery racks according to claim 9.
